

Spotify Music Analytics Dashboard

Project Overview

The Spotify Music Analytics Dashboard is a Power BI project developed to analyze music trends, artist performance, genre popularity, and the relationship between audio features and song popularity.

The objective of this project is to transform raw Spotify music data into meaningful insights using data cleaning, visualization, and analytical techniques.

Dataset Information

Dataset: Spotify Tracks Dataset

Total Records: 113,999 songs

Total Artists: 31,428

Total Genres: 114

Key Attributes:

- Track Name
 - Artist Name
 - Album Name
 - Popularity
 - Danceability
 - Energy
 - Valence
 - Tempo
 - Duration
 - Genre
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Project Objectives

The main objectives of this project were:

1. Identify the most popular genres.
2. Identify the most popular artists.
3. Determine the most popular songs.
4. Analyse the relationship between audio features and popularity.
5. Build an interactive dashboard for music trend analysis.

Data Cleaning Process

The following preprocessing steps were performed:

- Removed unnecessary serial number column.
 - Checked dataset structure and data quality.
 - Identified missing values.
 - Removed records containing incomplete track information.
 - Verified data types for all columns.
 - Loaded cleaned data into Power BI for analysis.
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Key Performance Indicators (KPIs)

The dashboard includes the following KPIs:

Total Songs

113,999

This indicates that the analysis is based on a large and diverse music collection.

Total Artists

31,428

This demonstrates a wide representation of artists across different music genres.

Total Genres

114

This diversity allows the comparison of popularity across multiple music categories.

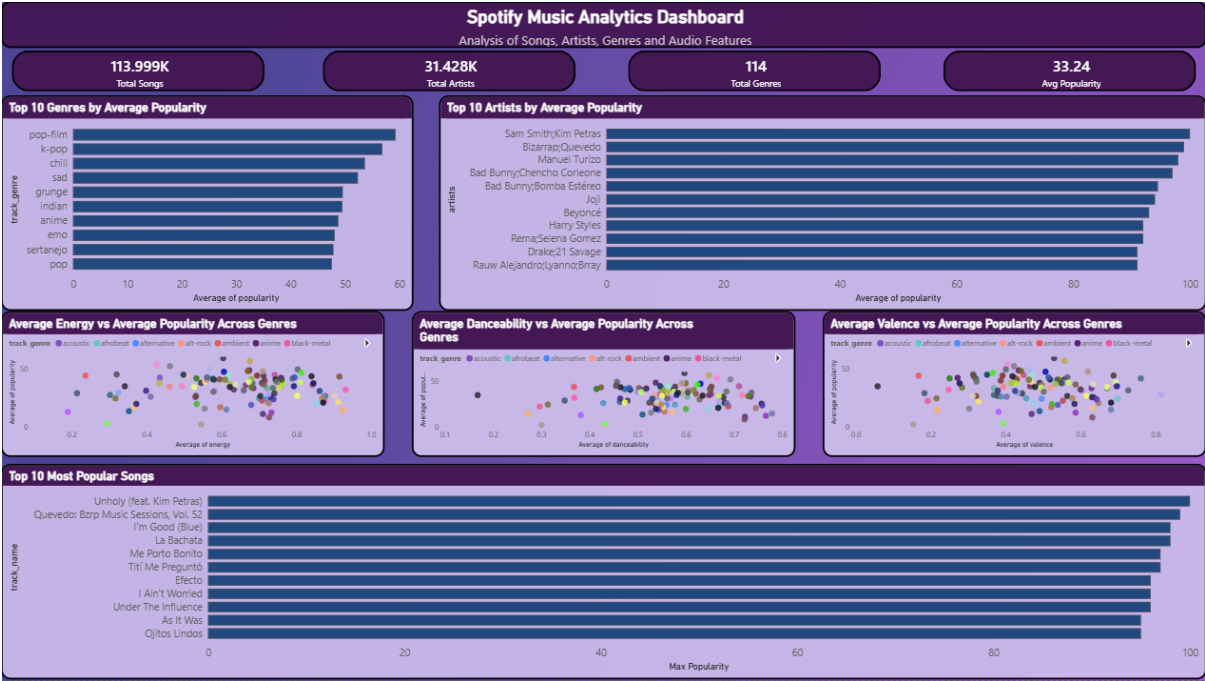
Average Popularity

33.24

This serves as a benchmark for comparing genres, artists, and songs

These KPIs provide a quick overview of the dataset.

Dashboard Analysis



Dashboard Overview

The dashboard is divided into three sections. The top section presents KPIs providing a summary of the dataset. The middle section focuses on genre, artist, and audio feature analysis. The bottom section highlights the most popular songs based on popularity scores.

1. Top Genres by Average Popularity

This visualization identifies genres with the highest average popularity.

Key Findings:

- Pop-film emerged as one of the most popular genres.
- K-pop demonstrated strong popularity.
- Chill and Sad genres also ranked highly.

2. Top Artists by Average Popularity

This chart highlights artists and artist collaborations with the highest average popularity scores.

Key Findings:

- Several globally recognized artists ranked among the top performers.
 - Artist collaborations showed particularly high popularity.
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3. Top 10 Most Popular Songs

This visualization identifies the most popular songs within the dataset.

Key Findings:

- A small number of tracks achieved exceptionally high popularity scores.
 - Popular songs significantly outperform average tracks.
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4. Energy vs Popularity Analysis

A scatter plot was used to study the relationship between average energy levels and popularity.

Observation:

- The scattered distribution of points indicates a weak correlation between energy and popularity.

Key Findings:

- Highly energetic genres are not always the most popular.
 - Energy alone does not strongly determine popularity.
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5. Danceability vs Popularity Analysis

This analysis investigates whether danceable songs tend to be more popular.

Observation:

- The clustering of points around moderate-to-high danceability values suggests that danceable tracks tend to achieve better popularity scores.

Key Findings:

- A moderate positive relationship was observed.
 - Genres with higher danceability often achieved better popularity scores.
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6. Valence vs Popularity Analysis

Valence measures the positivity or happiness of music.

Observation:

- The absence of a clear trend indicates that both positive and negative emotional tones can achieve audience success.

Key Findings:

- Both high-valence and low-valence genres achieved popularity.
 - Listener preferences are influenced by multiple factors beyond mood.
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Major Insights

1. Pop-film and K-pop genres showed the highest average popularity.
 2. Artist collaborations frequently appeared among top-performing artists.
 3. Danceability demonstrated a stronger relationship with popularity than energy.
 4. Popularity is influenced by multiple audio characteristics.
 5. Music preference patterns vary significantly across genres.
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Conclusion

The Spotify Music Analytics Dashboard successfully transformed raw music data into actionable insights through data visualization and analysis.

The project demonstrates the use of Power BI for:

- Data Cleaning
- KPI Development
- Data Visualization
- Trend Analysis
- Insight Generation

Overall, the analysis suggests that music popularity is influenced by a combination of genre, artist recognition, and audio characteristics. While attributes such as danceability show some relationship with popularity, no single factor alone guarantees success.

This dashboard can help users understand music trends, audience preferences, and the characteristics associated with popular songs.